Fatema Tuj Johora Faria

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Research Interests

Large Language Models, Vision-Language Models, AI Agents, Trustworthy AI, AI Security, and AI Safety.

Education

Ahsanullah University of Science and Technology (AUST) July 2019 – December 2023
B.Sc. in Computer Science and Engineering (CGPA: 3.302 / 4.00) *Dhaka, Bangladesh*

- **Undergraduate Thesis:** Investigated generative adversarial networks (CycleGAN and Pix2Pix) for potato disease image generation to address limited training data, and performed instance-level segmentation using Mask R-CNN.
- **Thesis Supervisor:** Prof. Dr. Mohammad Shafiul Alam

Publications

Workshops

- **Fatema Tuj Johora Faria**, Mukaffi B. Moin, Jubayer A. Mahmud, M. F. Mridha, and Md. A. Hossain. **"CLAIR-Fin: An Adversarial Multi-Agent Framework for Claim-Level Verification and Adaptive Debate in Cross-Modal Financial QA,"** *Proceedings of the 11th Workshop on Financial Technology and Natural Language Processing*, 2026. [FinNLP-2026 @ EMNLP-2026] Paper Code Dataset
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Jubayer Al Mahmud, M. F. Mridha, and Md. Alam Hossain. **"LabourCrew: A Multi-Agent RAG Framework for Trustworthy Adversarial Deliberation and Statutory Reasoning over Labour Law,"** *arXiv preprint arXiv:2609.27814*, 2026. [To be Submitted, Bangla Language Processing (BLP) Workshop @ NAACL 2027] Paper Code Dataset
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, M. F. Mridha, and Jubayer Al Mahmud. **"TeachMateGPT: A Multi-Agent Knowledge-Grounded Framework for Pedagogical Assessment Generation from Science Curriculum Materials,"** *arXiv preprint arXiv:2608.13708*, 2026. [To be Submitted, Bangla Language Processing (BLP) Workshop @ NAACL 2027] Paper Code
- **Fatema Tuj Johora Faria**, Mukaffi B. Moin, Md. M. Rahman, Khan M. Hasib, Jubayer A. Mahmud, and M. F. Mridha. **"Guiding Language Models to Be More Empathetic: Culturally Sensitive Mental Health Advice Generation Through Human-LLM Collaboration,"** *arXiv preprint arXiv:2607.23538*, 2026. [To be Submitted, Bangla Language Processing (BLP) Workshop @ NAACL 2027] Paper

Conferences

- Elias Hossain, Md Mehedi Hasan Nipu, **Fatema Tuj Johora Faria**, Shubhashis Roy Dipta, Maleeha Sheikh, Tasfia Nuzhat Ornee, and Niloofar Yousefi. **"ChannelGuard: From Borrowed Safety to Application-Controlled Defenses in Multi-Agent LLMs,"** *arXiv preprint arXiv:2607.19430*, 2026. [Submitted to NAACL 2027 Industry Track] Paper
- Elias Hossain, Md Mehedi Hasan Nipu, **Fatema Tuj Johora Faria**, Shubhashis Roy Dipta, and Ser-Nam Lim. **"When Supervision Fails: A Survey of Integrity Attacks and Defenses in LLM Alignment,"** [Submitted to NAACL 2027]
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Rabeya Islam Mumu, Md. Mahabubul Alam Abir, Abrar Nawar Alfy, and Mohammad Shafiul Alam. **"Motamot: A Dataset for Revealing the Supremacy of Large Language Models Over Transformer Models in Bengali Political Sentiment Analysis,"** *2024 IEEE Region 10 Symposium (TENSYMP)*, 2024, pp. 1–8. Paper Code Dataset

Journals

- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Md Mehedi Hasan Nipu, Khan Md Hasib, Md Rafiqul Islam, and Sami Azam. **"HEAL: A Multi-Agent Framework with Disagreement-Gated Validation for Reliable Text-to-SQL on Clinical Patient Records,"** 2026. [Manuscript in Preparation for Expert Systems with Applications] Project Details
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Ahmed Al Wase, Mehidi Ahmmed, Md. Rabius Sani, and Tashreef Muhammad. **"Vashantor: A Large-Scale Multilingual Benchmark Dataset for Automated Translation of Bangla Regional Dialects to Bangla Language,"** *arXiv preprint arXiv:2311.11142*, 2025. [Under Review in Language Resources and Evaluation] Paper Code Dataset
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Zayeed Hasan, Md. Arafat Alam Khandaker, Niful Islam, Khan Md Hasib, and M. F. Mridha. **"MultiBanFakeDetect: Integrating Advanced Fusion Techniques for Multimodal Detection of Bangla Fake News in Under-Resourced Contexts,"** *International Journal of Information Management Data Insights*, 5(2), 100347, 2025. Paper Code Dataset
- **Fatema Tuj Johora Faria**, Mukaffi Bin Moin, Busra Kamal Rafa, Swarnajit Saha, Md. Mahfuzur Rahman, Khan Md Hasib, and M. F. Mridha. **"BanglaCalamityMMD: A Comprehensive Benchmark Dataset for Multimodal Disaster Identification in the Low-Resource Bangla Language,"** *International Journal of Disaster Risk Reduction*, 130, 105800, 2025. Paper Code Dataset

Ongoing Research Projects

- Investigating and Mitigating Positional Bias in Medical Imaging through 2D Vision-Language Models and 3D Medical Image Segmentation Models

Professional Experience

Astha IT November 2025 – Present
AI Engineer II *Dhaka, Bangladesh*

Product: TalentX | *Python, LangGraph, LangChain, LangSmith, OpenAI, PostgreSQL, WebSocket, FastAPI, React, GCP Services, AWS Services*

- Directed end-to-end technical delivery of TalentX through stakeholder requirement analysis, feedback review, architectural decisions, proof-of-concept (PoC) validation, cross-functional coordination, and production deployment.
- Built a "Job Builder Agent" that combines PDF/DOCX parsing, field-level information extraction, and human-in-the-loop refinement to transform user requirements or uploaded documents into structured, editable job specifications.
- Designed and developed a "Job Fit Scoring Agent" that generates job-specific evaluation criteria and weights, assesses candidates against requirements, and delivers evidence-based evaluation reports to HR teams via email.
- Architected a "Talent Matching Agent" that converts HR queries into structured retrieval constraints and combines semantic vector search, metadata filtering, and relevance reranking to retrieve evidence-backed candidate recommendations from the CV bank.
- Automated candidate CV ingestion and synchronization through scheduled batch jobs, with content-change detection to reprocess updated CVs, refresh their vector embeddings, and remove outdated records to maintain accurate and up-to-date search results.
- Implemented a multi-tenant architecture with organization-level data isolation, user authentication, role-based permissions, and agent-level access controls to restrict data access and agent operations based on organizational boundaries and user privileges.
- Configured a "Candidate Communication Agent" to automate job application notifications, interview invitations, and individual or bulk rejection emails through HR-defined templates to streamline recruitment processes.
- Optimized large-scale CV processing through batch execution, parallel processing, and redundant LLM-call reduction, enabling the evaluation of 800–1,000 candidate profiles and **reducing processing time from 5 days to under 12 hours**.

Dexian (Bangladesh) Limited July 2025 – October 2025
Senior Application Developer *Dhaka, Bangladesh*

Project: ShareFlow Agent | *Python, LlamaIndex, Langfuse, Azure OpenAI, Microsoft Azure Services, PostgreSQL (pgvector), FastAPI, RAGAS*

- Architected a multi-agent RAG system integrating SharePoint and web sources via dynamic tool invocation, leveraging ReAct reasoning and source-aware context construction for knowledge-grounded responses.
- Developed a "Retrieval Agent" with ReAct planning, query reformulation, retrieval filtering, and custom guardrails to constrain responses to internal documents and reduce unsupported generation.
- Orchestrated an "OCR Agent" to extract content from diverse unstructured sources (scanned PDFs, images, and DOCX files) with rate limiting to support fault-tolerant performance.
- Integrated a "Web Browser Agent" with controlled web retrieval and instruction-aware source selection to augment knowledge-grounded responses with current external information when enterprise sources were insufficient.
- Implemented dynamic knowledge-base management with incremental SharePoint synchronization, file-level tracking, and memory updates, allowing users to add or remove knowledge sources without rebuilding the complete index.
- Designed session-aware conversational workflows with persistent history, context management, and session reset mechanisms, and integrated the system with Microsoft Bot Framework for deployment within Microsoft Teams.
- Mentored junior developers through system design reviews, code reviews, debugging, and technical knowledge sharing.
- Achieved an average **60–65% annual operational cost reduction** through agent and inference optimization for 80 Sales Managers with 50+ interactions per day.

Dexian (Bangladesh) Limited May 2024 – June 2025
Application Developer *Dhaka, Bangladesh*

Project: Org Info | *Python, LangGraph, LangChain, LangSmith, Azure OpenAI, Microsoft Azure Services, React, FastAPI, OpenCV, WebSocket*

- Co-led project delivery by analyzing stakeholder feedback, refining requirements, coordinating technical implementation, and managing continuous releases from development through production.
- Implemented an "OrgChart Extraction Agent" that analyzes uploaded organograms, identifies reporting relationships, reconstructs organizational hierarchies, and maps extracted roles to corresponding Bullhorn database records.
- Developed an "OrgQuery Agent" that converts natural language queries into optimized SQL (text-to-SQL), retrieves organizational hierarchy data from a relational database, and displays results through an interactive OrgChart interface.
- Architected an "OrgInfo Assistant" that converts user queries into SQL, executes them against a temporary organization-specific database to reduce load on the primary database, and translates query results into natural language.
- Created an "OrgActivity Summarization Agent" to generate 7-day summaries of organizational activities by extracting relevant notes on placements, submissions, and communication logs from the Bullhorn database.
- Designed an interactive React frontend with chat-based query input, hierarchical organization graph visualization, and WebSocket integration for real-time updates between the user interface and backend agents.
- Automated Bullhorn synchronization through scheduled data ingestion and incremental database updates, **cutting organizational hierarchy search time by 75%** by eliminating repeated direct queries against the source system.

Project: RFPMatcher | *Python, LlamaIndex, Langfuse, DeepEval. Azure OpenAI, Microsoft Azure Services, CouchDB, React, FastAPI, WebSocket*

- Architected a document-centric RAG pipeline with Chain-of-Thought reasoning and 12 specialized analytical modules to extract key information (e.g., project scope and deliverables) from Request for Proposal (RFP) documents.
- Orchestrated a master RFP database by processing historical responses with metadata (generated answers, summaries, win/loss labels, and question categories) and vector embeddings to support semantic search.
- Designed a "Past Experience Matcher" system that analyzes new RFP documents by extracting key client requirements, generating binary (Yes/No) and descriptive questions for each section, and querying a master database to identify relevant past experiences through semantic similarity matching.
- Developed a Table of Contents (TOC) generation workflow that analyzes extracted RFP key information against predefined section libraries to intelligently select and prioritize relevant sections based on project requirements.
- Implemented a "Proposal Writing Assistant" that generates initial content for each TOC section using extracted RFP information and relevant past experience, enabling iterative refinement through targeted chat interactions.
- Accelerated decision-making through predictive insights, with **manual review of 100+ pages reduced by 3–5 days** and Proposal Managers able to focus on strategic bid development.

Academic Reviewing

Academic Peer Reviewer 2025 – Present

Journals & Workshops

- **Journals:** Scientific Reports; Language Resources and Evaluation; Discover Mental Health; Cluster Computing
- **Workshop:** FinNLP-2026 @ EMNLP-2026

Technical Skills

- **Programming Languages:** Python (Pandas, NumPy, SciPy, Seaborn), Java, C++, JavaScript, TypeScript
- **Deep Learning Frameworks:** TensorFlow, Keras, PyTorch
- **LLM Application Frameworks:** LangChain, LangGraph, LlamaIndex, CrewAI, OpenAI Agent SDK, Claude Agent SDK
- **LLM Evaluation and Observability:** LangSmith, Langfuse, RAGAS, DeepEval
- **Vector Databases:** Milvus, ChromaDB, Qdrant, PostgreSQL (pgvector)
- **Cloud Services (Microsoft Azure):** Azure AI Foundry, Azure SQL Database, Azure Blob Storage, Azure App Service
- **Cloud Services (AWS):** Elastic Container Registry, Elastic Compute Cloud, Simple Storage Service, Simple Email Service
- **Cloud Services (GCP):** Cloud Storage, Cloud Run, Cloud SQL
- **Others:** Hugging Face, LLM Fine-tuning, OpenCV, PostgreSQL, React, Docker, Microsoft Bot Framework, WebSocket, Redis

Awards & Achievements

- Received a **50% registration fee scholarship** for research excellence and presentation of "*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*" at the 11th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2025).
- Secured a **Top 5 position among 50 posters** at the AUST Research Symposium 2023: An Intra-AUST Research Exhibition for undergraduate thesis research on "*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*".
- Recognized by **DEXIAN (Bangladesh) Limited** for **Excellence in Strategic Leadership, Agile Project Delivery, and Results-Driven Execution** in H1 2025.
- Led **Team AUSTPhoenix** to secure **7th place** in the **"Bengali Document Layout Analysis Competition"** at DL Sprint 2.0 - BUET CSE Fest.

Tutoring Experience

Private Academic Tutor 2020 – 2023

Mathematics, Physics & Chemistry, Grades 11–12 *Dhaka, Bangladesh*

- Provided one-on-one instruction in conceptual understanding, problem-solving, and examination preparation.
- Created customized lesson plans and practice materials aligned with individual learning needs, supported by regular assessment and constructive feedback.